A One-Scale Positive-Kernel $K + R$ Basin Criterion

Abstract

We derive an exact composition law for a comparison-controlled reference channel and a state-weighted residual channel. Combining the local-cylinder comparison with Gaussian concentration yields a nonempty numerical budget inside the sharp nonlinear radius. The remaining task is to realize the two subkernels for the complete nonlinear $SU(2)$ conditional measure.

1 Composition theorem

Let K_0, K, R be positive bad subkernels on a retained-state space, and let $V > 0$.

Theorem 1. *If*

$$K \le e^{\ell} K_0, \qquad K_0 V \le kV, \qquad RV \le rV,$$

then

$$(K + R)V \le (e^{\ell} k + r)V.$$

In particular, the Stage 19 connected-cluster theorem applies whenever

$$e^{\ell} k + r < e^{-\alpha} \Delta^{-2}.$$

Proof. Positivity permits the three inequalities to be applied without cancellation: $KV \le e^{\ell} K_0 V \le e^{\ell} kV$, and addition gives the claim. □

2 Feasible local radius

Let $\delta(r)$ be the Stage 21 local Gaussian union upper bound and let $\ell(r)$ be the Stage 23 comparison loss per extension. We select r so that

$$e^{\ell(r)} \delta(r) = \frac{q_{\mathrm{target}}}{2}, \qquad q_{\mathrm{target}} = 0.002430277501841998.$$

Monotonicity on the certified interval and bisection give

$$r_* = 0.00614955590043087.$$

This is only 53.69% of the sharp nonlinear maximum 0.011453785959974176.

Quantity	value
$\ell(r_*)$	$2.40009685 \cdot 10^{-5}$
reference k	0.00121510958676
corrected $e^{\ell} k$	0.00121513875092
residual budget r	0.00121513875092
combined ceiling	0.00243027750184
cluster branching ratio	0.870222533796

3 Frontier

The budget polytope is nonempty. This is stronger than merely observing overlap of local radii: it allocates explicit room to both reference and charged residual channels. However, the complete nonlinear conditional measure has not yet been decomposed into physical K and R satisfying the hypotheses. The residual weighted partition numerator is the next estimate.